



Light-weight 3DCNN for DeepFakes, FaceSwap and Face2Face facial forgery detection

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Abstract

Detecting facial forgery in real time application such as video conferencing, netbanking, verification of identity etc. is a major concern. Highly accurate facial forgery detectors will be incompatible with limited computation and storage space devices like smartphones, tablets, PCs due to their deep complex architectures. Therefore, we propose a light weight 3D convolutional neural networks (3DCNN) to detect popular facial forgeries namely, DeepFakes, Face2Face and FaceSwap. The proposed 3DCNN is a five layered architecture with only 2.69M trainable parameters. The proposed network learns spatio-temporal features to detect forged videos. Further, to understand the detection process of 3DCNN, activation maps are studied in detail. The popular FaceForensic++ dataset is employed to train and test the proposed method. In addition, the effectiveness of the proposed method is evaluated and compared with state-of-the-art methods in terms of both, number of trainable parameters and binary detection accuracy.

Keywords 3DCNN · Face forensics

1 Introduction

With the evolving deep learning techniques and current work from home scenarios, video conferencing for business purpose, facial verification, virtual social connects have become common. These deep learning techniques have made life easier, however, increasing malicious use of it has led to an increase in facial forgeries. Thus, designing efficient and real time compatible facial forgery detectors is of prime importance.

The most popular facial forgeries are namely, DeepFakes, Face2Face and FaceSwap. DeepFakes and FaceSwap are identity based forgery while Face2Face is an expression based forgery. The facial manipulations are even employed using feature rejuvenation, aging and anthropometric simulations [4] and [5]. In literature, the researchers have employed

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various tools to detect these facial manipulations. In [11], the authors have rigorously exploited steganalysis techniques to detect forgery. In [17], patch classification is employed, while in [7], image processing algorithms are used to design facial forgery detectors. Even, complex 2D and 3D CNNs are employed and trained on huge facial forgery dataset to detect DeepFakes, FaceSwap and Face2Face. The Xception Net [6] is a 36 layered architecture, which comprises of depthwise separable convolutional layers with residual connections. The Xception Net works efficiently in classifying large classes dataset. It is trained on FaceForensic++ dataset [18] to detect facial forgeries. Similarly, Wang et al. in [26], trained and fine tuned 3D ResNet [12], the excellent action recognition network, on the famous FaceForensic++ dataset [18]. Moreover, generative adversarial networks(GAEL-Net [3]) are also used in designing robust facial manipulation detectors. Therefore, in order to achieve high detection accuracy, researchers have begun to design more and more complex architectures.

These highly accurate facial forgery detectors are designed at the cost of network size and complexity. A highly efficient Xception network [6], when benchmark on FaceForensics++ dataset, achieves high DeepFake binary detection accuracy. However, it consists of almost 22M trainable parameters. Similarly, network based on pre-trained VGG-16 [21], ResNet [13], 3D ResNets [12], optical flow based method [1] are computationally expensive networks. Due to their large size and high computation, these efficient networks are not compatible with limited computation and space devices, such as smartphones, personal laptops and tablets. Whereas, lightweight deep learning models are easier to train and require less overheads to update when deployed on smartphones, personal laptops and tablets.

Consider a video conferencing application, which is popularly and widely used on PCs, laptops and smartphones. Many important business and personal discussions are done through video conferencing. Let's consider two types of video conferencing applications, the first application consists of a small and lightweight facial manipulation detector while, the second does not have any facial forgery detector. Therefore, if during the video conference, someone with a forged face attends the conference, then only in the first type of application fraud will be checked/detected. Thus, real world applications, highly demand lightweight models. Therefore, this paper proposes a light weight 3D convolutional neural networks (3DCNN) for facial manipulation detection. The proposed method does not employ complex and computational expensive networks like FlowNets [9] or PWCNet [22] to extract temporal features of the test video. Rather exploit 3DCNN in a novel manner to extract both spatial and temporal features. Furthermore, the proposed 3DCNN is designed with less number of trainable parameters, which again results in a simple and light weight architecture.

The main contributions of this work are as follows:

1. To design a light weight facial forgery detector using a novel, 3D convolutional neural network (3DCNN).
2. The 3D convolutional neural network (3DCNN) is employed in such a manner that it initially learns spatial features and later learns the combined spatio-temporal features of a face.
3. Finally, the proposed method is compared with MobileNetV2 [19], XceptionNet [6] 3D ResNet [26] and fCNN [16].

The paper is organized as follows: Section 1 describes the introduction of facial manipulation and related detection methods, Section 2 elaborates the motivation behind the proposed method, Section 3 describes the working of the proposed method, Section 4 contains a description of the dataset used in the experiment and the metrics for analyzing the proposed method, Section 5 contains activation maps, results and discussions, and Section 6 concludes the paper.

2 Motivation

Nowadays, the research emphasis is on designing small and efficient networks so that computer vision applications can be brought to limited computation platforms, such as smartphones, tablets, personal laptops, etc. In MobileNetV1 [14] and MobileNetV2 [19] author reduced computation in initial layers by using depth wise separable convolution. The lightweight MobileNet is applied for object detection, face recognition and other computer vision applications. Further, it is compared with exiting models like VGG-16 [21], GoogleNet [23] etc. for COCO object detection dataset. It is observed that MobileNet achieves equivalent accuracy with reduced computation cost. Similarly, when compared with popular FaceNet [20], the number of trainable parameters are reduced by half while multiple-adds are reduced by eight times. However, the accuracy also drops by almost 4%. The same has been observed for deep fired convnets [27] and SqueezeNet [15], where the authors designed the network to achieve equivalent or 2%-3% lower accuracy at significantly lower computational cost.

Therefore, it can be inferred from the above that compromising with a slight decrease in accuracy can significantly reduce the number of trained parameters. The resulting reduced parameter model can be applied to real world applications. This prompted us to design a computationally efficient and light weight 3DCNN network.

The proposed 3DCNN is extracting both spatial and temporal features without using complex and computationally expensive optical flow methods. Rather employed simple image processing tool, i.e x and y direction gradients. Rather simple image processing tools, i.e. x and y direction gradients are employed. These x and y direction gradients present strong candidates in the facial manipulation detection process. The Fig. 1 describes in detail how the gradients in DeepFakes, FaceSwap and Face2Face faces are changing as compared to the pristine face. The faces in Fig. 1 are extracted from pristine video number “216.mp4” and their corresponding DeepFakes [8], Face2Face [24] and FaceSwap [10] videos of Face-Forensic++ dataset [18]. The Fig. 1a-d represents faces from DeepFakes, ‘DF’, Face2Face, ‘F2F’, FaceSwap, ‘FS’ and Pristine, ‘Pris’ videos. The two gradients G_x and G_y are computed for the above four faces. The gradient G_x and G_y represent vertical edges and horizontal edges in a face, respectively. The Fig. 1e-h represents G_x face gradients while the Fig. 1i-l represents G_y face gradients of DF, F2F, FS and Pris, respectively.

On comparing gradients of DeepFakes with gradients of pristine, i.e comparing Fig. 1e with h and i with l, it is visualized that there are more vertical and horizontal edges around the cheeks and eyebrows. Similarly, in case of Face2Face, observe Fig. 1f and j with respect to (h) and (l), abrupt edges can be seen around lips and forehead regions as this is a face reenactment forgery. Likewise, in the case of FaceSwap, Fig. 1g and k, region around the chin and mouth are exhibiting artifacts. Thus, high gradient values are observed in forged facial regions as compared to pristine. This unique feature inspired us to employ gradients in 3DCNN.

3 Proposed method

The proposed method is divided into two stages as shown in Fig. 2. The first is pre-processing stage, where the video is converted into 4 depth matrix. This 4D matrix comprises of two consecutive frames and its x and y direction gradients. The computed 4D matrix is given as an input to 3DCNN in second stage. The second stage is employed to detect the forged face using the proposed 3DCNN.

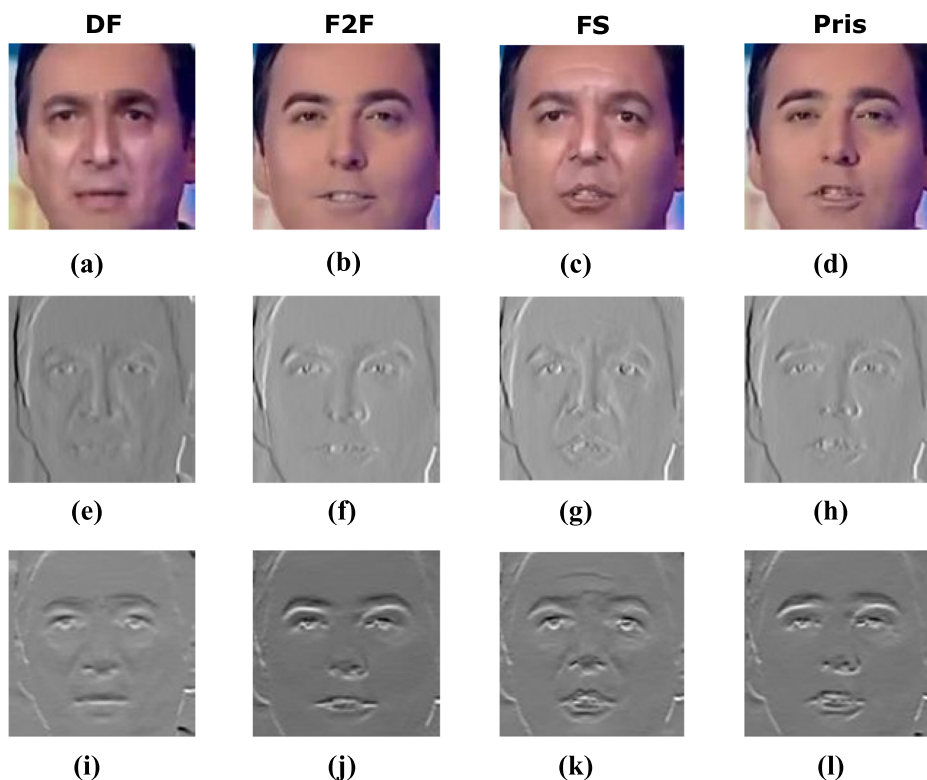


Fig. 1 (a)–(d) represents DeepFakes, Face2Face, FaceSwap and Pristine faces respectively. (e)–(h) represents their corresponding x-direction gradient. (i)–(l) represents their corresponding y-direction gradient

3.1 Pre-processing stage

In this stage, the test video is processed through different operations. First, the test video is split into RGB frames using ffmpeg toolbox. Then the target faces are extracted from the RGB frames using Viola-Jones method [25]. The extracted RGB faces are converted into gray scale faces. Further, the Sobel method is employed to compute x and y direction gradients of the test face. In this method gradients are computed such that the edges in the

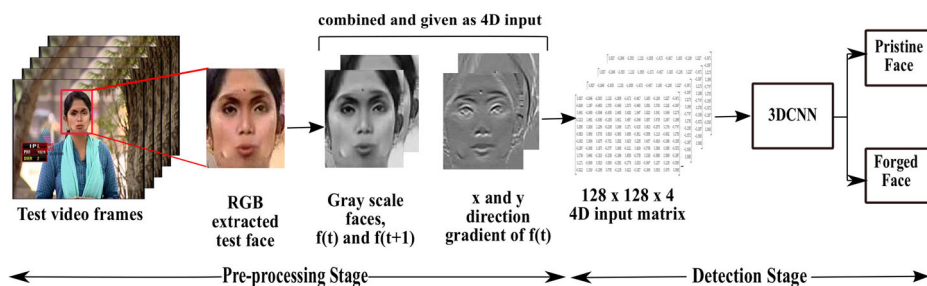


Fig. 2 Block Diagram of the proposed method

face are highlighted. The forged face gradient will either consist of abrupt edges or smooth regions. Let t be the current frame, and $f(t)$ represents the test face, then x direction, G_x^t and y direction, G_y^t gradients are calculated by convolving 3×3 filters with $f(t)$ as follows,

$$G_x^t = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} * f(t) \quad (1)$$

$$G_y^t = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * f(t) \quad (2)$$

where, $*$ denotes 2D convolution operation. The calculated G_x^t and G_y^t are fed as an input to 3DCNN. Further, to extract temporal information between the frames, the two consecutive frames, $f(t)$ and $f(t+1)$ are also fed to network. Thus, forming a 4D input matrix corresponding to each test face, $f(t)$.

3.2 Detection stage

The 4D face matrix computed in pre-processing stage of proposed method is fed as an input to 3DCNN in detection stage. The 4D face matrix comprises of two consecutive frames, horizontal gradient and vertical gradient, i.e. $[f(t), f(t+1), G_x^t, G_y^t]$. The 3DCNN process the input and learn the key features to classify the test face into pristine or forged.

The conventional CNN is a 2 dimensional convolutional neural network, where features are extracted from xy plane or spatial domain of input. However, to include features from other domains as well, a 3 dimensional convolutional neural network is required. In 2D CNN, the 2D kernels are convolved with input, thus output is a 2D. While in 3DCNN the 3D kernels are convolved with an input cube formed by stacking three input layers. Therefore, taking into account the combined effect of the xy plane and depth of input and then generating a 2D output.

Let us consider that there are d number of channels (depth) in the input of size $M \times N$, i.e. input size is $(M \times N \times d)$. The kernel height, weight and length are $(P \times Q \times R)$, respectively. Then, the output feature map at position (s, t, u) after convolution of input cube with 3D kernel is given as,

$$output_{stu} = \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} \sum_{r=0}^{R-1} w^{pqr} i^{(s+p)(t+q)(u+r)} \quad (3)$$

where, w^{pqr} is the weight of 3D kernel at (p, q, r) position and for p, q, r equal to zero, i^{stu} is the u th input at position (s, t) .

The proposed 3DCNN is designed with five 3D convolutional layers, as shown in Fig. 3. The green squares in Fig. 3 represent 2D kernels and the red cubes represent 3D kernels. All convolutional layers are followed by an activation layer of ReLu function. The batch normalization is employed in the proposed network, to accelerate the training process as well as to mitigate the effect of unstable gradients in the model. Initially, in proposed 3DCNN, the $(3 \times 3 \times 1)$ size kernel is selected to learn the individual effect of gray scale frames and gradients. Later, as we proceed towards third layer combined effect of consecutive frames and gradients is taken into consideration. Thus, the kernel size of $(3 \times 3 \times 2)$ is selected for third, fourth and fifth convolutional layers. The detailed description of 3DCNN is discussed in Table 1 of model summary.

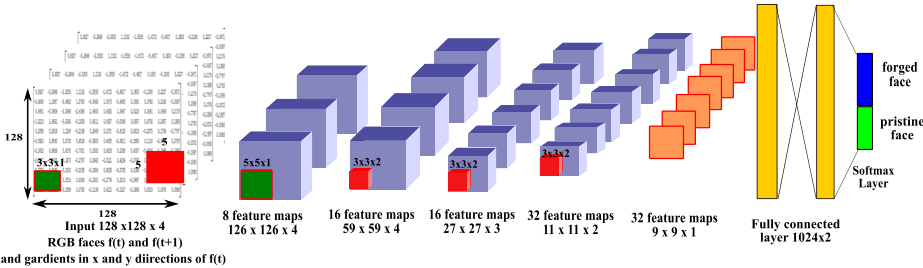


Fig. 3 The proposed 3DCNN architecture

4 Dataset

The proposed method is trained and tested on FaceForensics++ dataset [18]. The FaceForensics++ dataset consists of different types of facial forgeries, namely DeepFakes, Face2Face, FaceSwap, and Neural textures. It contains 1000 pristine videos and their corresponding 4000 forged videos, 1000 each from four different facial forgery techniques. The dataset

Table 1 The model summary of proposed 3DCNN

Layer (type)	Output shape	Trainable parameters
Input Layer	128 x 128 x 4	0
conv3d_1 (Conv3D)	126 x 126 x 4 x 8	80
activation_1 (Activation)	126 x 126 x 4 x 8	0
batch_normalization_1(batch)	126 x 126 x 4 x 8	32
max_pooling_3d_1	63 x 63 x 4 x 8	0
conv3d_2 (Conv3D)	59 x 59 x 4 x 16	3216
activation_2 (Activation)	59 x 59 x 4 x 16	0
batch_normalization_2(batch)	59 x 59 x 4 x 16	64
max_pooling_3d_2	29 x 29 x 4 x 16	0
conv3d_3 (Conv3D)	27 x 27 x 3 x 16	4624
activation_3 (Activation)	27 x 27 x 3 x 16	0
batch_normalization_3(batch)	27 x 27 x 3 x 16	64
max_pooling_3d_3	13 x 13 x 3 x 16	0
conv3d_4 (Conv3D)	11 x 11 x 2 x 32	9248
activation_4 (Activation)	11 x 11 x 2 x 32	0
batch_normalization_4(batch)	11 x 11 x 2 x 32	128
conv3d_5 (Conv3D)	9 x 9 x 1 x 32	18464
activation_5 (Activation)	9 x 9 x 1 x 32	0
batch_normalization_5(batch)	9 x 9 x 1 x 32	128
flatten_1	2592	0
dense_1	1024	2655232
dropout_2	1024	0
dense_2	2	2050
Total trainable parameters		2,693,122

is generated from random collection of youtube videos, comprises of both the genders, i.e. male and female. The dataset covered VGA (480p), HD (720p) and FHD (1080p) resolution videos. The dataset also consists of different compression quality videos, i.e. c0, c23 and c40. The c0 videos are uncompressed videos while c23 and c40 are compressed videos with H.264 codec. The high quality videos are generated with constant quantization rate of 23, and low quality videos are generated with constant quantization rate of 40. Thus, the FaceForensics++ dataset is the most generic dataset of its kind.

All the experiments in the proposed method are mainly conducted on three facial forgeries, namely DeepFakes, Face2Face and FaceSwap. DeepFakes are deep learning based forgery while Face2Face and FaceSwap are computer generated graphics based forgery. The experiments are performed on all the compression qualities available in the dataset, i.e. c0, c23, c40. In order to efficiently train the 3DCNN to differentiate between forged and pristine face, the dataset is divided into three groups, namely training, validation and testing with 700, 150 and 150 videos, respectively. The 3DCNN model is even cross validate to generate a robust model. The videos are split into frames using ffmpeg. These frames are used for further processing. Detailed statics about the number of frames used in each class are discussed in Table 2.

The performance of the proposed method is evaluated using binary detection accuracy, Ac and number of trainable parameters used in the model. Ac , measures the number of correct test faces detected from total number of test faces. The number of trainable parameters measures the complexity of the model. The lesser the number of trainable parameters, the simpler the model.

$$Ac = \frac{Correct_{faces}}{total_{faces}} * 100 \quad (4)$$

5 Experimental discussion

In this section, the robustness of the proposed 3DCNN is studied. Experiments conducted for different facial forgery scenarios considering all possible combinations of facial forgery techniques and their compression ratios. Further, the activation maps for c40 DF, FS and F2F is studied. The comparison of proposed 3DCNN with Amerini et al. method [1] and Wang et al. [26] 3D ResNet method is studied in detail. The proposed method is also evaluated in terms of training parameters to understand its computation complexity. All the experiments are performed on Xeon E5 CPU, 3.50Hz, RAM 64GB and Xeon E5 CPU, 3.60 Hz RAM 16GB. The MATLAB is used to prep-process the FaceForensic++ dataset. The training and testing of the proposed method is done on keras and tensorflow frameworks(Ubuntu 18.04 3LTS).

Table 2 The cardinality of all classes in the three datasets, namely DeepFake, FaceSwap and Face2Face.(training set, 'Tr', validation set, 'val' and testing set, 'Test')

Dataset	DeepFake		Face2Face		FaceSwap		Pristine	
	Tr+val	Test	Tr+val	Test	Tr+val	Test	Tr+val	Test
c0	29609	4966	29804	4949	23586	4046	29738	5056
c23	30452	5098	31020	5135	24516	4133	30787	5207
c40	30407	5108	31058	5114	24516	4192	30804	5152

The proposed 3DCNN is designed to classify between forged and pristine face. The model is trained end-to-end for different forgery techniques. The model is cross validated to avoid over-fitting. The model hyper-parameters are selected through intensive experimentation and best model is selected to conduct the experiments. The model is trained for the three different forgeries namely, DeepFakes, 'DF', Face2Face, 'F2F', FaceSwap, 'FS' for all the compression ratios (c0, c23, c40). This results in nine trained models. These nine trained models are further tested on the unseen 150 test videos. The validation and testing accuracy of these nine models are discussed in detail in Table 3. On analyzing the table, it is observed that uncompressed forged video detection is better than compressed detection in all the facial forgery techniques(DF, F2F, FS). The forged video detection accuracy falls for c40 compression while for c0 and c23 compressions the detection accuracy is almost same. On visualizing the table, from facial forgery technique prospective, it is seen that the DeepFakes detection is almost 7% better than F2F and 3% better than FS for c40 compression. Therefore, it can be inferred from the Table 3, the proposed method is best in detecting DeepFake facial forgeries.

5.1 Activation map visualization and interpretation

The activations maps are helpful in visualizing the discriminative regions used by a CNN to classify the test image [28]. Therefore, to understand the classification process of proposed 3DCNN, a detailed investigation is conducted for all the three facial forgeries. To this end, randomly fifty faces are selected from the unseen c40 test FaceForensics++ dataset [18] for all the three forgeries, i.e. DeepFakes, FaceSwap and Face2Face and their pristine counterparts. Further, their activation maps are calculated at the last convolutional layer of 3DCNN of size 9x9x32. Among 32 filters randomly four are chosen. And for these four filters, mean of all fifty activation maps are calculated and presented using colormaps in Figs. 4, 5 and 6. In colormaps, the dark red color represents strongly activated facial region and dark blue represents the weakly activated facial region.

The Fig. 4a-d present DeepFakes mean activation maps and Fig. 4e-h present pristine mean activation maps. As discussed in motivation Section 2 in DeepFake faces, artifacts are observed around face boundaries, forehead and cheeks region. Similar observations can be visualized in activation maps of Fig. 4. The Fig. 4a and b show strong activation around face boundary and cheeks region while its counterpart pristine activation maps Fig. 4e and f show weak activation for these areas. Similarly, Fig. 4c and d present strongly activated forehead region while in the case of pristine Fig. 4g and h, weakly activated region is observed.

The Fig. 5a-d present FaceSwap mean activation maps and Fig. 5e-h present pristine mean activation maps. The FaceSwap forged faces are either blurred or extremely smooth. Thus details of face are missing in a forged face, especially in nose, chin, lips and cheeks

Table 3 The validation and testing accuracy of the proposed method for DeepFakes (DF), Face2Face(F2F) and FaceSwap(FS) of FaceForensics++ dataset [18]

	c0		c23		c40	
	Validation	Testing	Validation	Testing	Validation	Testing
DF	0.9906	0.9200	0.9908	0.9163	0.9853	0.9099
F2F	0.9713	0.8673	0.9507	0.8785	0.9363	0.8348
FS	0.9648	0.9053	0.9601	0.8695	0.9526	0.8759

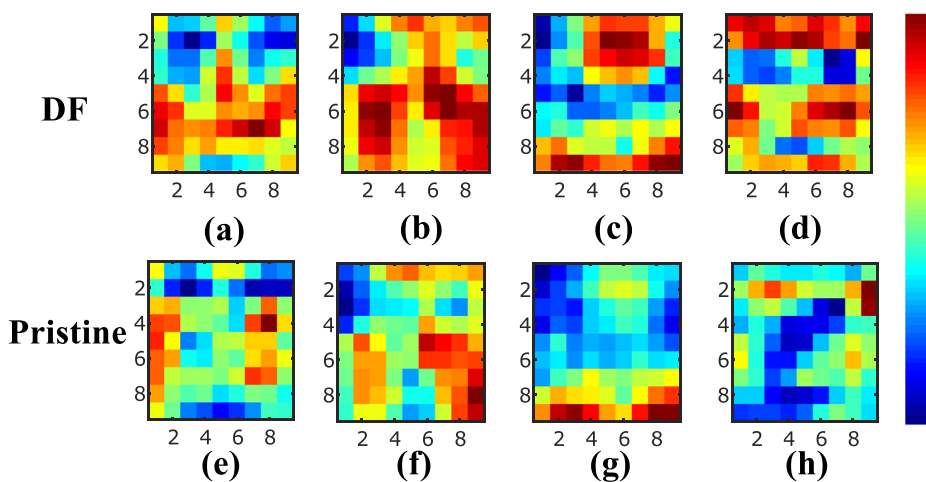


Fig. 4 The mean activation maps of four randomly selected filters of last convolutional layer of proposed 3DCNN of size 9x9 for c40 DeepFakes and their corresponding pristine faces of test FaceForensics++ dataset [18]

region. In Fig. 5e, g and h, the strong activations are observed in cheeks, lips and chin region, while weak activation is observed in its forged counterpart, i.e. Fig. 5a, c and d. This shows that, these particular filters are learning details of the pristine face, which are otherwise absent in FaceSwap faces due to forgery. Similarly, the Fig. 5b presents strong activation in nose region while Fig. 5f presents weak activation. In the same manner, Face2Face activation maps are observed in Fig. 6. The Face2Face forgery is an expression based forgery, therefore the details of expressions are either missed or deliberately impainted in forged faces. The face2Face mean activation maps are presenting strong activation in forged regions as seen in Fig. 6c and d. While weak activation in the details of a facial expressions, especially in regions near lips and mouth as seen in Fig. 6a and b.

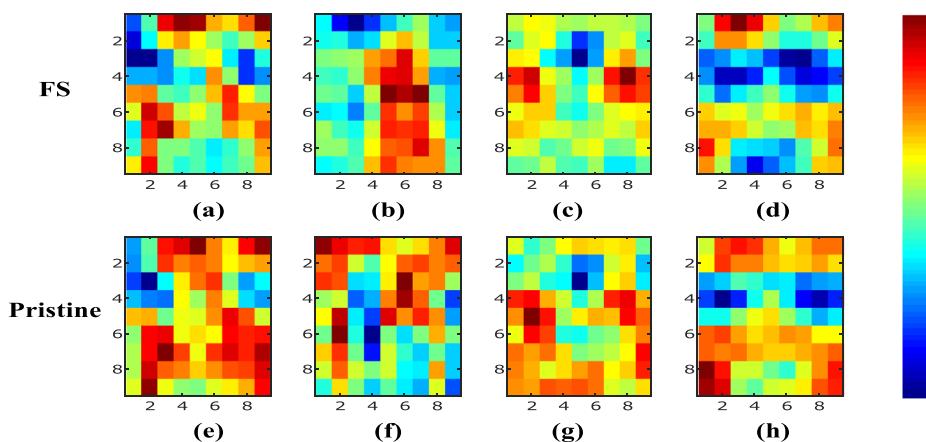


Fig. 5 The mean activation maps of four randomly selected filters of last convolutional layer of proposed 3DCNN of size 9x9 for c40 FaceSwap and their corresponding pristine faces of test FaceForensics++ dataset [18]

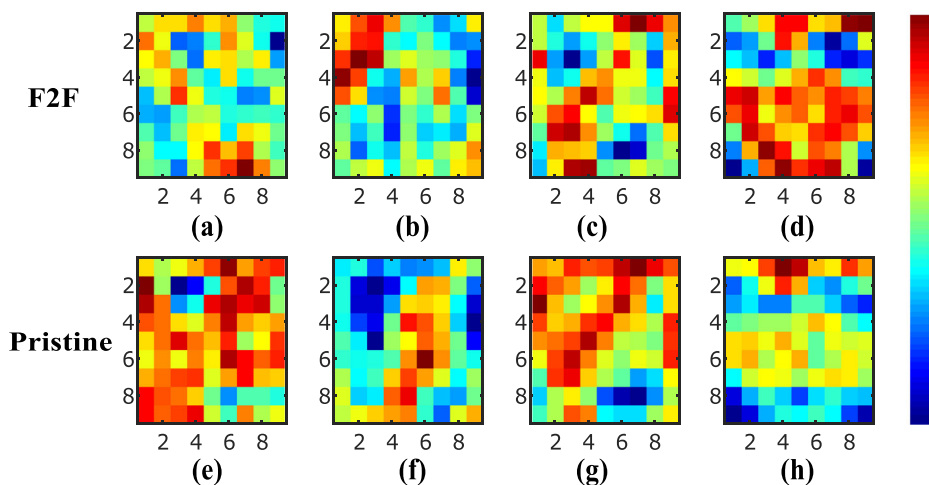


Fig. 6 The mean activation maps of four randomly selected filters of last convolutional layer of proposed 3DCNN of size 9x9 for c40 Face2Face and their corresponding pristine faces of test FaceForensics++ dataset [18]

Thus, it is inferred from the activation map analysis, that different filters of proposed 3DCNN are learning discriminative features of face and forgery artifacts to classify the test face.

5.2 Comparative analysis

The proposed method is compared with Amerini et al. [1] method. The Amerini et al. [1] method is an optical flow based forgery detection method. In this method the optical flow is calculated using PWC-Net [22]. The computed optical flow is fed into the pre-trained CNN. The author tested their proposed method for VGG-16 [21] and ResNet50 [13] pre-trained models. The author conducted experiments only on c23 Face2Face FaceForensics++ dataset [18]. In order to compare the proposed method with Amerini et al. [1] method, the new test video dataset of 120 unseen of F2F c23 videos is generated as per the details provided by the author. The Table 4 lists the testing accuracy of the proposed method and Amerini et al. [1] method for both the pre-trained models, i.e. VGG16 and ResNet50. It is observed that, the proposed method outperforms the Amerini et al. [1] method. The detection accuracy of the proposed method is almost 4% and 10% better than Amerini et al. [1] method for VGG16 and ResNet50, respectively. The proposed method is also compared in terms of computation complexity with Amerini et al. [1] method. As discussed in previous section the computational complexity is measured in terms of trainable parameters. The Table 5 discusses the number of trainable parameters in the different detection methods. VGG16

Table 4 The testing accuracy of proposed method and Amerini et al. [1] method for c23 Face2Face FaceForensics++ dataset [18]

	c23 F2F
Amerini et al. method(VGG16)	81.61
Amerini et al. method (ResNet 50)	75.46
proposed method	85.96

Table 5 The trainable parameters of proposed method and Amerini et al. [1] method for c23 Face2Face FaceForensics++ dataset [18]

	Trainable parameters (in Millions)
Amerini et al. method(VGG16)	138.35M
Amerini et al. method (ResNet 50)	25.56M
proposed method	2.69M

[21] and ResNet50 [13] method are very complex as compared to the proposed method. The proposed method uses about 10 times less number of trainable parameters, i.e. only 2.69M.

The FaceForensic++ dataset [18] is the most famous and challenging facial manipulation dataset. The robustness of the FaceForensic++ dataset can be evaluated using objective inpainting metrics(OIM) [2], forgery detection accuracy of detector etc. In [18], the authors conducted an experiment to understand how humans perform in detecting facially manipulated videos (facial forgery). And they observed that, humans struggle to differentiate between forged and pristine faces for low quality videos (c40). Therefore, facial forgery detection under c40 manipulation scenario is highly challenging. Thus, the robustness of proposed method is evaluated and compared with other facial manipulation detectors for c40 quality videos.

The Table 6 lists the comparison of proposed method with Wang et al. [26] 3D ResNet method, Xception Net [6], MobileNet V2 [19] and fCNN [16]. The comparison is done in terms of binary detection accuracy and number of trainable parameters required for each method. To this end, a new test dataset of 140 videos each from three manipulations, i.e. DF, F2F, FS of c40 compression ratio is generated as per the details provided by the author of [26]. The binary detection accuracy of proposed 3DCNN and 3D ResNet [26] method is almost the same for DeepFakes and FaceSwap, but lower in the case of Face2Face. While the trainable parameters of proposed 3DCNN method are about 12 times less than 3D ResNet [26]. Moreover, the Face2Face detection accuracy of proposed 3DCNN method and XceptionNet [6] are similar, but again the trainable parameters of proposed method are about 10 times less than XceptionNet [6]. However, the proposed 3DCNN has 0.47 Millions more trainable parameters than MobileNetV2 [19], but the detection accuracy of the proposed method is higher than MobileNetV2 [19] for all the three cases (DF, F2F and FS). Therefore, it can inferred from the Table 6 that, detection accuracy of 3D ResNet [26] and Xception-Net [6] can be achieved with lower network complexity (number of trainable parameters). Hence, the results in Tables 4, 5 and 6 prove the efficacy of the proposed method in terms of its detection accuracy and network complexity.

Table 6 The binary detection accuracy (in%) and number of trainable parameters evaluated for c40 (DF, F2F and FS) test videos of FaceForensics++ dataset [18], for different facial forgery detection methods

	DF	F2F	FS	Trainable Parameters(in Millions)
fCNN [16]	78.75	66.68	67.76	44.25 M
3D ResNet [26]	91.81	89.60	88.75	33.23 M
Xception [6]	89.93	83.04	88.82	20.81 M
MobileNet V2 [19]	87.25	81.53	85.33	2.22 M
proposed method	91.03	83.72	87.33	2.69 M

6 Conclusion

In the event of detecting forged facial videos, authors have begun to design complex architectures that are incompatible with real time applications. Therefore, a light weight network is proposed to work with real time applications, such as laptops, smartphones etc. The proposed method employs simple image processing tools such as gradients to pre-process the face and further fed them to a novel five layered 3D convolutional neural network (3DCNN) to classify between forged and pristine faces. The activation maps of proposed 3DCNN are visualized and studied in detail. The activation maps make inferences about the discriminative features of face and forgery artifacts learned during training of proposed 3DCNN.

The proposed method is compared with Amerini et al. [1], Wang et al. [26], XceptionNet [6], MobileNetV2 [19] and fCNN [16] methods in terms of binary detection accuracy and trainable parameters. Comparative analysis shows that the proposed method achieves better accuracy with only 2.69M of trainable parameters, which is about 10 times less than Wang et al. [26] and XceptionNet [6].

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Declarations

Conflict of Interests Aditi Kohli and Abhinav Gupta declare that they have no conflict of interest. Aditi Kohli receives research grant, file No. 09/1132(0010)/2019 EMR-I, from Council of Scientific and Industrial Research, India (CSIR).

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